Portfolio health snapshot

Synthetic demo portfolio • data 2019-01-01 to 2026-09-22 • 6 holdings, 2,016 trading days

● 7 within range ● 2 to watch ● 1 outside range

PORTFOLIO VALUE
$100,000
6 holdings, long-only

1-DAY 99% VaR (FHS)
● 2.45%
$2,449; 95% CI 2.27%–2.68%

1-DAY 99% EXPECTED SHORTFALL
2.82%
mean loss beyond VaR; $2,819

ANNUALISED VOLATILITY
20.3%
trailing 1y: 25.4%

MARKET BETA
● 1.14
factor R^2 86%

EFFECTIVE POSITIONS (RISK)
● 5.7
vs 5.5 by weight

WORST HISTORICAL STRESS
● −15.2%
Synthetic window B

RISK MODEL (BACKTEST)
● 1.12%
garch_normal breaches vs 1% expected

Health checklist

	Check	Value	Rule	Reading
●	Risk model validity	garch_normal: 1.12% breaches	coverage & ES tests $p > 0.05$	Best model passes 2/2 tests; Basel zone green over 1,516 days.
●	Daily VaR vs limit	2.45% (49% of limit)	limit 5.0%; amber at 80%	1-day 99% FHS: a one-in-a-hundred day costs about $2,449.
●	Largest risk contributor	SYN03: 23.4% of VaR	amber ≥20%, red ≥30%	24.4% of capital carries 23.4% of risk.
●	Diversification (effective positions)	5.7 by risk / 5.5 by weight	amber <10, red <5	Number of equal-sized independent bets the book behaves like.
●	Sector concentration	Technology: 33.1% of VaR	amber ≥40%, red ≥60%	36.1% of capital.
●	Market beta	1.14	green 0.8–1.2, amber 0.5–1.5	Factors explain 85.9% of variance.
●	Worst historical stress	−15.2%	amber ≤−20%, red ≤−35%	Synthetic window B replayed at today's weights.
●	1-year max drawdown	−28.0%	amber ≤−15%, red ≤−25%	Full-sample max drawdown −63.0%.
●	Volatility regime	0.79× average	FHS / historical VaR; amber ≥1.1	Current conditions are calmer than the sample average.
●	Data quality (proxied history)	0.0% of capital	weight in names >50% proxied; amber ≥10%	Mostly-proxied: none.

Takeaways. Largest risk contributor: SYN03 is 24.4% of capital and 23.4% of VaR; the book behaves like 5.7 independent positions. Worst historical replay: Synthetic window B at −15.2%. Most damaging hypothetical: Equity −20% at −22.8%.

Thresholds are defaults in `risklib.output.pdf.DEFAULT_THRESHOLDS`; override them under `pdf: thresholds:` in the portfolio config. Colours flag where to look, not what to do.

Risk summary

How much could be lost (1 day, 99%)

Method	VaR	95% CI	ES	VaR $
Parametric normal	2.97%		3.40%	$2,972
Historical	3.09%	2.83%–3.30%	3.53%	$3,092
Filtered historical (GARCH)	2.45%	2.27%–2.68%	2.82%	$2,449
MC t-copula + empirical	3.07%		3.50%	$3,072

10-day 99% VaR (FHS): 8.43%.

Where the risk comes from

Holding	Weight	Share of VaR
SYN03	24.4%	23.4%
SYN01	22.6%	20.3%
SYN05	16.2%	17.5%
SYN04	12.3%	13.2%
SYN02	11.0%	12.9%

Factor exposures

Factor	Beta
MKT	1.14
SIZE	−0.01
VALUE	0.12

Stock-specific share of variance: 14.1%.

Sectors

Sector	Weight	Share of VaR
Technology	36.1%	33.1%
Healthcare	24.4%	23.4%
Industrials	16.2%	17.5%
Energy	12.3%	13.2%
Financials	11.0%	12.9%

Historical replays (worst eight)

Scenario	Return
Synthetic window B	−15.2%
Synthetic window A	−6.2%

Hypothetical shocks (worst six)

Scenario	Return
Equity −20%	−22.8%
Value +15%	1.8%

Can the VaR be trusted? (out-of-sample backtest, 1,516 days, 1% expected)

Model	Breaches	Rate	Kupiec p	Indep. p	ES Z_2	ES p	Zone
normal	23	1.52%	0.060	0.359	0.59	0.021	yellow
historical	21	1.39%	0.154	0.294	0.43	0.054	green
garch_normal	17	1.12%	0.641	0.183	0.13	0.312	green
fhs	20	1.32%	0.234	0.264	0.32	0.110	green

$p > 0.05$ = not rejected. Bold = model selected for the headline VaR.

Allocation reference (walk-forward, out of sample)

Portfolio	Return	Vol	Sharpe	Max DD
Current weights	-6.2%	20.5%	-0.21	−63.0%
Risk parity (Ledoit-Wolf)	-8.6%	20.2%	-0.34	−64.7%

Detail

The complete report, section by section. Numbers are exactly those in `report.md`.

1. Portfolio

ticker	weight %	sector	risk % (VaR)
SYN00	13.45	Technology	12.78
SYN01	22.62	Technology	20.28
SYN02	11.02	Financials	12.92
SYN03	24.44	Healthcare	23.38
SYN04	12.30	Energy	13.18
SYN05	16.16	Industrials	17.46

Concentration

Effective number of positions by weight **5.5**, by risk **5.7** (1/Herfindahl). Top-1 holding 24.4% of weight and 23.4% of VaR; top-5 89.0% / 87.2%.

Sectors

sector	weight %	risk %	n holdings
Technology	36.1	33.1	2
Healthcare	24.4	23.4	1
Industrials	16.2	17.5	1
Energy	12.3	13.2	1
Financials	11.0	12.9	1

Risk profile (daily-rebalanced at today's weights)

index	daily rebalanced	buy and hold
ann_vol	0.2028	0.2095
max_drawdown	−0.6302	−0.6503
skew	0.0020	0.0040
kurtosis	0.3014	0.4118
hit_rate	0.5064	0.5055
n_obs	2016	2016

Trailing-window performance

These are the figures to compare with a benchmark. The full-sample row, and all return statistics, describe today's weights applied to the past — hindsight, not expectation (see appendix).

window	ann return	ann vol	sharpe	max drawdown	n obs	start
1y	0.2990	0.2535	1.1587	−0.2795	252	2025-10-06
3y	−0.0521	0.2069	−0.1553	−0.4417	756	2023-10-31
5y	−0.0721	0.2103	−0.2506	−0.6302	1260	2021-11-24
full	−0.0413	0.2028	−0.1066	−0.6302	2016	2019-01-01

2. Factor model (OLS, Newey-West HAC, 7 lags)

Portfolio variance explained by factors: **85.9%**. Factors: MKT=MKT, SIZE=SIZE, VALUE=VALUE. Proxied holdings' betas use real data only (`real_only` column).

Portfolio exposures

factor	exposure
MKT	1.142
SIZE	−0.006
VALUE	0.122

Variance decomposition (annualized)

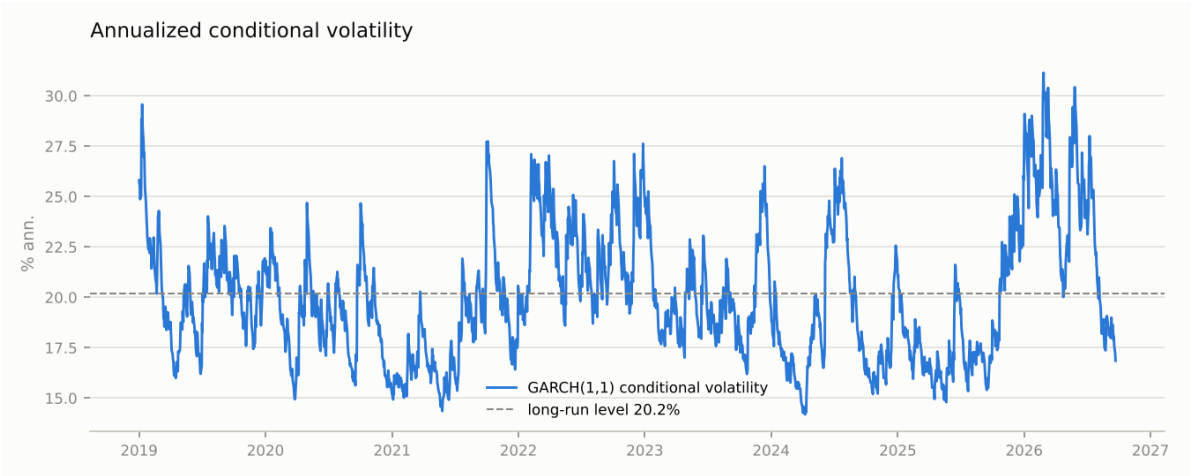
source	variance	vol	pct of total
MKT	0.0349	0.1869	85.3773
SIZE	0.0000	0.0011	0.0032
VALUE	0.0002	0.0139	0.4744
idiosyncratic	0.0058	0.0761	14.1452
total	0.0409	0.2023	100.0000

Per-asset betas

ticker	MKT	SIZE	VALUE	r2	idio vol ann	n obs	real only
SYN00	1.054	−0.031	0.404	0.512	0.174	2016	yes
SYN01	0.953	0.110	0.554	0.463	0.179	2016	yes
SYN02	1.223	0.090	0.010	0.579	0.171	2016	yes
SYN03	1.335	−0.606	0.092	0.613	0.181	2016	yes
SYN04	1.220	0.102	−0.546	0.586	0.173	2016	yes
SYN05	1.073	0.616	−0.085	0.513	0.180	2016	yes

3. Volatility (portfolio returns)

GARCH(1,1) conditional volatility is **16.9%** annualized against a long-run level of 20.2%; shocks decay with a half-life of 27 days (persistence 0.975).



Out-of-sample forecast comparison (rolling refit)

model	qlike	mse	dm vs EWMA(0.94)	dm pvalue
EWMA(0.94)	1.362	6.791e-08		
GARCH(1,1)	1.366	6.799e-08	0.4833	0.6289
GJR-GARCH	1.367	6.818e-08	0.6179	0.5367

Lower QLIKE is a better variance forecast; `dm_pvalue` < 0.05 means the difference from the first model is statistically significant (Diebold-Mariano).

4. VaR / ES (fraction of portfolio value; losses positive)

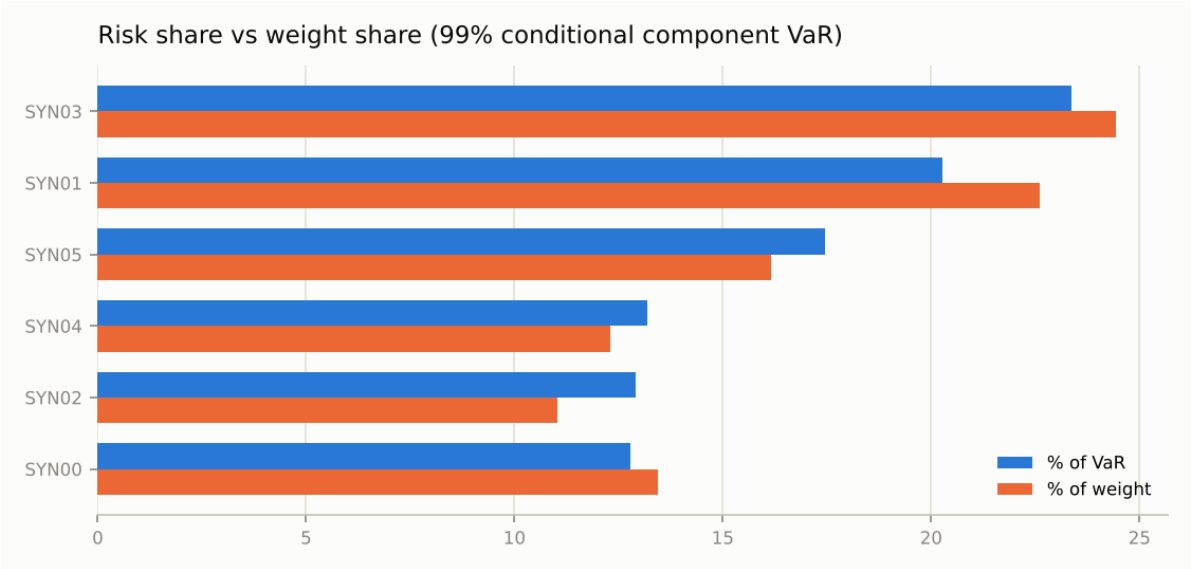
`VaR_lo`/`VaR_hi` are bootstrap 95% confidence intervals (historical: resampled days; FHS: resampled standardized residuals, GARCH parameters fixed). Multi-day historical rows are $\sqrt{h}$-scaled from the 1-day estimate. Monte Carlo uses the Ledoit-Wolf covariance. Methods and confidence levels are set by `risk.methods` and `risk.alphas`.

method	horizon	alpha	VaR	ES	VaR lo	VaR hi	ES lo	ES hi	VaR $	ES $
Parametric normal	1	0.9900	0.0297	0.0340					2971.6018	3404.4588
Historical	1	0.9900	0.0309	0.0353	0.0283	0.0330	0.0324	0.0376	3092.4950	3531.7916
Filtered historical (GARCH)	1	0.9900	0.0245	0.0282	0.0227	0.0268	0.0261	0.0302	2449.4392	2818.7157
MC t-copula + empirical	1	0.9900	0.0307	0.0350					3072.4055	3499.8635
Parametric normal	1	0.9750	0.0250	0.0299					2503.5949	2986.2340
Historical	1	0.9750	0.0256	0.0308	0.0235	0.0272	0.0289	0.0325	2556.1240	3078.1479
Filtered historical (GARCH)	1	0.9750	0.0211	0.0247	0.0199	0.0225	0.0234	0.0263	2111.3822	2468.3413
MC t-copula + empirical	1	0.9750	0.0245	0.0304					2453.1003	3039.8881
Parametric normal	10	0.9900	0.0940	0.1077					9397.0301	10765.8440
Historical ($\sqrt{10}$-scaled from 1-day)	10	0.9900	0.0978	0.1117					9779.3278	11168.5057
Filtered historical (GARCH)	10	0.9900	0.0843	0.0985					8425.1252	9852.3453
MC t-copula + empirical	10	0.9900	0.0888	0.0999					8882.4323	9987.0700
Parametric normal	10	0.9750	0.0792	0.0944					7917.0621	9443.3010
Historical ($\sqrt{10}$-scaled from 1-day)	10	0.9750	0.0808	0.0973					8083.1740	9733.9585
Filtered historical (GARCH)	10	0.9750	0.0714	0.0855					7137.7844	8553.3947
MC t-copula + empirical	10	0.9750	0.0755	0.0887					7547.5025	8868.9741

Component VaR 99% 1-day (Euler, EWMA(0.94) conditional covariance)

ticker	weight	marginal var	component var	pct of var	standalone var
SYN00	0.1345	0.0220	0.0030	12.7832	0.0042
SYN01	0.2262	0.0208	0.0047	20.2819	0.0070
SYN02	0.1102	0.0272	0.0030	12.9208	0.0037
SYN03	0.2444	0.0222	0.0054	23.3771	0.0077
SYN04	0.1230	0.0249	0.0031	13.1817	0.0049
SYN05	0.1616	0.0251	0.0041	17.4552	0.0062

Diversification benefit: 31.2% of the sum of standalone VaRs. The covariance is conditional, so the attribution reflects today's volatility like the FHS headline.



5. Stress tests

Historical replay

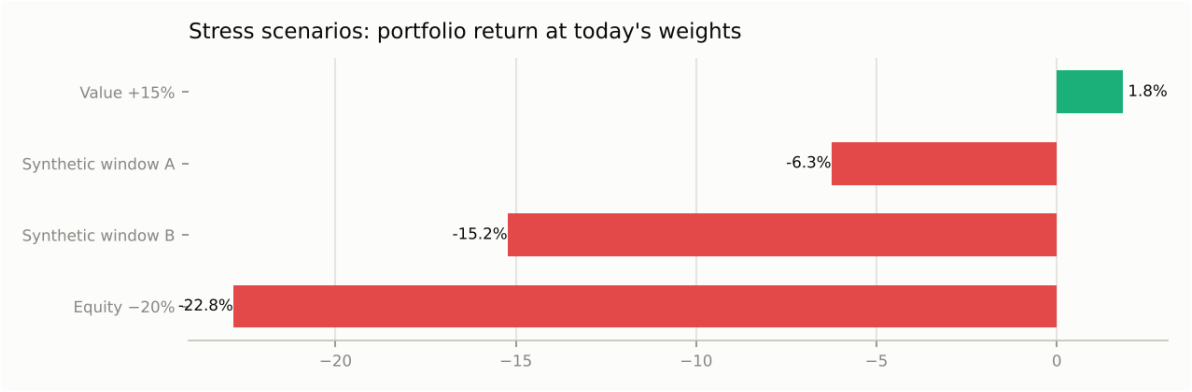
scenario	start	end	portfolio return	direct	backfilled	proxied	missing	factors missing
Synthetic window A	2020-03-02	2020-04-15	−0.0625	6	0	0	0	
Synthetic window B	2023-01-03	2023-03-31	−0.1523	6	0	0	0	

Today's weights applied to each window. Per holding, the window return comes from its own prices (`direct`), its proxy-backfilled history (`backfilled`), its factor betas times the factor returns over the window (`proxied`, idiosyncratic part zero), or is unavailable (`missing`, contributes 0). `factors_missing` lists factors whose ETFs did not trade in the window and so were left out of the `proxied` mapping. Price history for the replays is downloaded back to `stress.history_start`, independently of the estimation window.

Hypothetical factor shocks

scenario	portfolio return	shock	ignored factors
Equity −20%	−0.2284	MKT -20%	
Value +15%	0.0184	VALUE +15%	

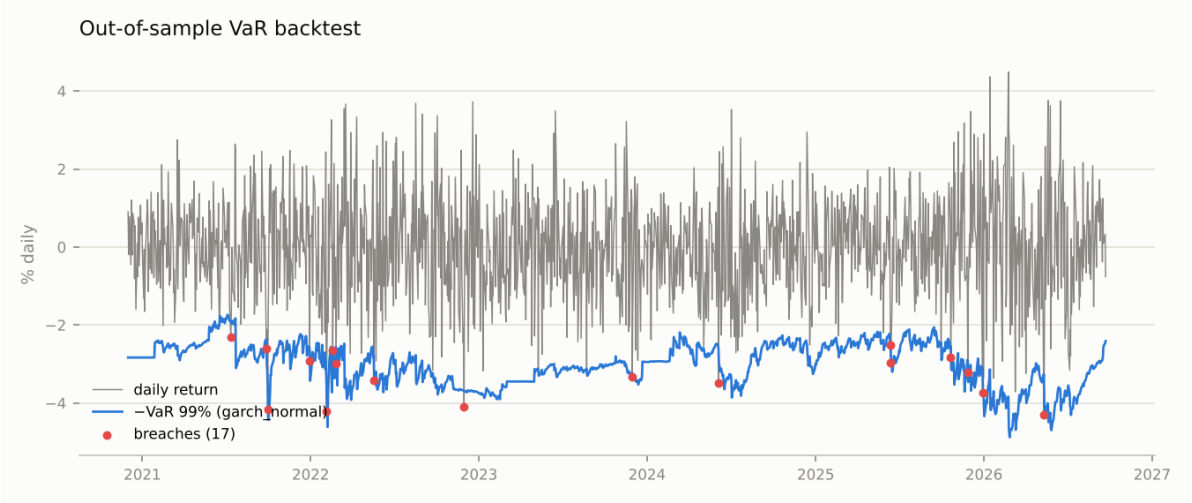
Rate shocks are mapped through equities the way 2022 transmitted them (per +100bp: MKT -10%, VALUE +7%); see `stress.RATE_SHOCK_PER_100BP`.



6. VaR backtest (99%, 1516 out-of-sample days, window 500)

method	n days	breaches	expected	breach rate %	kupiec p	christoffersen p	cc p	traffic light 250d	traffic light full	es Z2	es p	avg VaR %	max loss %
normal	1516	23	15.1600	1.5172	0.0601	0.3592	0.1121	red	yellow	0.5858	0.0210	2.8915	4.3030
historical	1516	21	15.1600	1.3852	0.1543	0.2943	0.2092	yellow	green	0.4344	0.0545	2.9208	4.3030
garch_normal	1516	17	15.1600	1.1214	0.6413	0.1828	0.3694	green	green	0.1254	0.3125	3.0060	4.3030
fhs	1516	20	15.1600	1.3193	0.2337	0.2640	0.2637	green	green	0.3224	0.1095	3.0308	4.3030

Best model by coverage among those passing conditional coverage: **garch_normal**. `garch_normal` uses a normal quantile on a GARCH σ, so comparing it with `fhs` (same σ, empirical tails) isolates the tail assumption. Models are set by `backtest.methods`. `es_Z2` > 0 means realized losses beyond VaR exceeded ES; `es_p` is the one-sided bootstrap probability. `traffic_light_250d` is the Basel zone on the last 250 days only; `traffic_light_full` applies the same binomial thresholds to the whole backtest.



7. Allocation reference

Walk-forward out-of-sample (window 500, rebalance every 126 days)

portfolio	ann return	ann vol	sharpe	max drawdown	calmar	avg turnover
sample/min_variance	−0.0981	0.1990	−0.4194	−0.6489	−0.1512	0.1660
sample/risk_parity	−0.0857	0.2019	−0.3427	−0.6474	−0.1324	0.0228
ledoit_wolf/min_variance	−0.0982	0.1990	−0.4197	−0.6490	−0.1513	0.1630
ledoit_wolf/risk_parity	−0.0857	0.2019	−0.3426	−0.6474	−0.1323	0.0226
equal_weight	−0.0831	0.2031	−0.3258	−0.6472	−0.1285	0.0000
current	−0.0619	0.2046	−0.2102	−0.6302	−0.0983	0.0000

Min-variance and risk parity need no expected-return estimate; max-Sharpe and target-vol allocators remain in `risklib.models.optimize` but are not reported because their out-of-sample results are dominated by return-estimation error.



Appendix: full-sample return statistics (hindsight)

Today's weights applied to the whole history. Because the weights are on stocks that have already performed, these figures overstate what the portfolio could have been expected to earn; they are kept for completeness only.

index	daily rebalanced	buy and hold
ann_return	−0.0413	−0.0282
sharpe	−0.1066	−0.0316
sortino	−0.1496	−0.0445
calmar	−0.0655	−0.0433
start	2019-01-01	2019-01-01
end	2026-09-22	2026-09-22

Generated by risklib. Every table is reproducible from `data/*.csv` with `data.source: csv`.